Solutions to Chapter 7 Exercises

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

Phase-plane sketches are described in words (equilibria, eigen-directions, orientation of the flow). For a potential system $\ddot{x} + P'(x) = 0$ I use the energy $E = \frac{1}{2}\dot{x}^2 + P(x)$, whose level curves are the orbits.

Exercise 7.1

$\ddot{x} + kx = 0$.

(a) With $y = \dot{x}$: $\dot{x} = y, \quad \dot{y} = -kx$.

(b) For $k > 0$ (Hooke's law) the origin is a center. The orbits are the ellipses $kx^2 + y^2 = c$ (semi-axes $\sqrt{c/k}$ in x and $\sqrt{c}$ in y), traversed clockwise with period $2\pi/\sqrt{k}$. (For $k < 0$ the origin is a saddle with eigenvalues $\pm\sqrt{|k|}$ and hyperbolic orbits; for $k = 0$ every point of the x-axis is an equilibrium and the orbits are horizontal lines $y = \text{const}$.)

Exercise 7.2

$\ddot{x} + 3\dot{x} - 4x = 0$.

(a) $\dot{x} = y, \quad \dot{y} = 4x - 3y$.

(b) The matrix $\begin{pmatrix} 0 & 1 \\ 4 & -3 \end{pmatrix}$ has $\lambda^2 + 3\lambda - 4 = 0$, so $\lambda = 1, -4$: the origin is a **saddle**. The unstable direction is the line $y = x$ (eigenvector $(1, 1)$, flow away from 0) and the stable direction is $y = -4x$ (eigenvector $(1, -4)$, flow toward 0). Other orbits are hyperbola-like curves, coming in along the stable line and leaving along the unstable line. The general solution is $x = c_1 e^t + c_2 e^{-4t}$.

Exercise 7.3

$\dot{x} = 2x - y$, $\dot{y} = x^2 + 4y$. Equilibria: $y = 2x$ and $x^2 + 8x = 0$, so $(0, 0)$ and $(-8, -16)$. The Jacobian is $\begin{pmatrix} 2 & -1 \\ 2x & 4 \end{pmatrix}$.

- $(0,0)$: $\begin{pmatrix} 2 & -1 \\ 0 & 4 \end{pmatrix}$ has eigenvalues $2, 4$: a source, **unstable**.
- $(-8,-16)$: $\begin{pmatrix} 2 & -1 \\ -16 & 4 \end{pmatrix}$ has $\lambda^2 - 6\lambda - 8 = 0$, so $\lambda = 3 \pm \sqrt{17}$ (one positive, one negative): a saddle, **unstable**.

Neither equilibrium is stable.

Exercise 7.4

$\dot{x} = a(x-b)(x-c)$, $b < c$.

(a) The equilibria are $x = b$ and $x = c$. The sign of $\dot{x}$ is the sign of a for $x < b$ and $x > c$, and the sign of $-a$ for $b < x < c$.

- $a > 0$: b is a sink and c is a source. If $x_0 < c$ then $x(t) \to b$. If $x_0 = c$ the solution is constant. If $x_0 > c$ then $x(t) \to +\infty$ (in finite time, see (c)).
- $a < 0$: reversed. c is a sink and b a source. If $x_0 > b$ then $x(t) \to c$. If $x_0 = b$ the solution is constant. If $x_0 < b$ then $x(t) \to -\infty$.
- $a = 0$: $\dot{x} = 0$, every solution is constant.

(b) Separating variables with $\frac{1}{(x-b)(x-c)} = \frac{1}{b-c}\left(\frac{1}{x-b} - \frac{1}{x-c}\right)$ gives

$$\frac{1}{b-c}\ln\left|\frac{x-b}{x-c}\right| = at + K, \qquad \text{so} \qquad \frac{x-b}{x-c} = \frac{x_0-b}{x_0-c}\,e^{-a(c-b)t}.$$

Solving for x:

$$x(t) = \frac{b(x_0-c) - c(x_0-b)e^{-a(c-b)t}}{(x_0-c) - (x_0-b)e^{-a(c-b)t}}.$$

At $t = 0$ this equals x_0. For $a > 0$, as $t \to \infty$ the exponential tends to 0 and $x(t) \to b$ (for $x_0 \neq c$); for $a < 0$ it tends to c (for $x_0 \neq b$). This agrees with (a).

(c) The denominator vanishes when $e^{-a(c-b)t} = \frac{x_0-c}{x_0-b}$. For $a > 0$ and $x_0 > c$ (or $a < 0$ and $x_0 < b$) this ratio lies in the range that the decreasing (resp. increasing) exponential reaches at the positive time

$$T = \frac{1}{a(c-b)}\ln\frac{x_0-b}{x_0-c},$$

so $x(t) \to \pm\infty$ as $t \to T^-$. The largest interval is $[0, T)$. This does **not** contradict the existence theorem: Theorem 7.14 only guarantees a solution on some (possibly short) interval around t_0, not for all time, and $f(x) = a(x-b)(x-c)$ grows quadratically.

(d) If $b = c$, then $\dot{x} = a(x-b)^2$ and $x(t) = b + \dfrac{x_0-b}{1-a(x_0-b)t}$. The equilibrium b is semistable. For $a > 0$: $x_0 < b$ gives $x \to b$, and $x_0 > b$ blows up at $t = \frac{1}{a(x_0-b)}$. For $a < 0$ the directions are reversed.

(e) Since $(b-x)(c-x) = (x-b)(x-c)$, the model is the same equation with $x_0 = 0$. Physically $a > 0$, concentrations $b, c \geq 0$ and $0 \leq x \leq \min(b, c)$, so only $a > 0$ with $x_0 = 0$ (below the smaller of b, c) is relevant. The cases $a < 0$ (a negative rate constant), $x_0 > c$ or $x_0 < b$ beyond the reactant concentrations, and blow-up are not physical. The outcome is $x(t) \to b$: the reaction proceeds until the limiting reactant Y (smaller concentration b) is used up. If $b = c$ the approach to b is algebraic, $x(t) = b\big(1 - \frac{1}{1+abt}\big)$.

Exercise 7.5

(a) $\begin{pmatrix} 3 & -1 \\ 2 & 4 \end{pmatrix}$: $\operatorname{tr} = 7$, $\det = 14$, $\lambda = \frac{7 \pm i\sqrt{7}}{2}$. The origin is a **spiral source**; orbits spiral outward counterclockwise (at $(1, 0)$ the velocity is $(3, 2)$).

(b) $\begin{pmatrix} -2 & 3 \\ 7 & -6 \end{pmatrix}$: $\lambda^2 + 8\lambda - 9 = 0$, $\lambda = 1, -9$. A **saddle**: unstable direction $(1, 1)$, stable direction $(3, -7)$.

(c) The (x, y) block $\begin{pmatrix} 3 & -1 \\ 1 & 3 \end{pmatrix}$ has $\lambda = 3 \pm i$ and $\dot{z} = -2z$. In polar form $r = r_0 e^{3t}$, $\theta = \theta_0 + t$, $z = z_0 e^{-2t}$. The origin is a saddle-focus: orbits spiral outward (counterclockwise) in the xy-directions while being pulled toward the plane $z = 0$. The z-axis is the stable manifold and the xy-plane is the unstable manifold.

Exercise 7.6

Attracting but not stable. In polar coordinates let $\dot{r} = r(1 - r)$, $\dot{\theta} = 1 - \cos\theta$. The point $(r, \theta) = (1, 0)$ is an equilibrium. Since $\dot{\theta} \geq 0$ with $\dot{\theta} = 0$ only at $\theta \equiv 0 \pmod{2\pi}$, every orbit with $r > 0$ has $\theta \to 2\pi \equiv 0$ and $r \to 1$, so $(1, 0)$ attracts every point except the origin. It is not stable: a point at angle $\theta = \varepsilon > 0$ must travel all the way around the circle to $\theta = 2\pi$, passing through $(-1, 0)$, a distance 2 from the equilibrium, however small ε is.

Stable but not attracting. For $\dot{x} = y$, $\dot{y} = -x$ (the oscillator), the origin is stable (orbits are circles that stay as close to 0 as we like), but no orbit tends to it.

Exercise 7.7

Let $\dot{v} = f(v)$ with f not depending on t (and C^1). Suppose solutions v and w pass through the same point p: $v(t_1) = p = w(t_2)$. Then $u(t) = w(t + t_2 - t_1)$ is also a solution (the equation is autonomous), and $u(t_1) = p = v(t_1)$. By uniqueness (Theorem 7.14), $u \equiv v$, so w is a time-shift of v and the two trace out the same orbit. Hence distinct orbits never cross.

Exercise 7.8

$\ddot{x} + x + x^3 = 0$.

(a) Take $E(x,\dot{x}) = \frac{1}{2}\dot{x}^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4$. Then $E(0,0) = 0$ and $E > 0$ elsewhere. Moreover

$$\dot{E} = \dot{x}\ddot{x} + x\dot{x} + x^3\dot{x} = \dot{x}(\ddot{x} + x + x^3) = 0 \leq 0,$$

so E is a Lyapunov function (not strict), and the origin is stable.

(b) $\dot{x} = y$, $\dot{y} = -x - x^3$. The only equilibrium is $(0,0)$ since $x(1+x^2) = 0$ forces $x = 0$. E is conserved, $\nabla E = (x + x^3,\, y)$ vanishes only at the origin, and $E \to \infty$ at infinity, so the level curves $\frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4 = c$ are nested closed curves around the origin. All orbits are periodic, traversed clockwise, and the origin is a (nonlinear) center.

Exercise 7.9

Write $\dot{x} = y$, $\dot{y} = -x - x^3 - y$ and use $E = \frac{1}{2}y^2 + \frac{1}{2}x^2 + \frac{1}{4}x^4$. Then

$$\dot{E} = y(-x - x^3 - y) + (x + x^3)y = -y^2 \leq 0.$$

E is positive definite and radially unbounded, so each sublevel set $W = \{E \leq c\}$ is compact and forward invariant. On $Q = \{\dot{E} = 0\} = \{y = 0\}$, an orbit staying in Q has $\dot{y} = 0$, so $x + x^3 = 0$, i.e. $x = 0$: the only invariant set in Q is the origin. By Barbashin–LaSalle (Corollary 7.26), every orbit starting in W tends to $(0,0)$. Since c is arbitrary, the origin is globally asymptotically stable.

Exercise 7.10

The system is $\dot{x} = y$, $\dot{y} = -P'(x)$, with equilibria $(x,0)$ at the critical points of P, and orbits on the level curves of $E = \frac{1}{2}y^2 + P(x)$.

(a) P has minima at x_1 and x_2 and a local maximum x_m between them. The equilibria $(x_1,0)$ and $(x_2,0)$ are **centers** (stable, not asymptotically stable), surrounded by closed orbits. $(x_m,0)$ is a **saddle** (unstable). The energy level $E = P(x_m)$ is a figure-eight-shaped separatrix with a homoclinic loop around each center. Below this level the orbits are small closed curves around one well; above it they are larger closed curves enclosing both wells.

(b) P has a minimum at x_1 and an inflection with $P'(x_2) = 0$ (so $P' \geq 0$ on both sides of x_2). $(x_1,0)$ is a center. At $(x_2,0)$ the linearization $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is degenerate, and the level curve $E = P(x_2)$ has a **cusp** there. It consists of two branches that approach $(x_2,0)$ in infinite time (as $t \to \pm\infty$), forming a loop around x_1. Orbits inside this loop are closed curves around x_1; orbits outside are larger closed curves around it. $(x_2,0)$ is **unstable**, because a point just to the right of it has energy above $P(x_2)$ and travels around the entire outer loop.

Exercise 7.11

$\dot{x} = y$, $\dot{y} = -P'(x) - by$, with $b > 0$. Along orbits $\dot{E} = -by^2 \leq 0$, so energy is lost and all orbits converge to equilibria. The equilibria are the same as in 7.10.

- **(a)** $(x_1, 0)$ and $(x_2, 0)$ are now **asymptotically stable** (spiral sinks if b is small, nodes if $b^2 > 4P''$), and $(x_m, 0)$ is a saddle. Every orbit tends to $(x_1, 0)$ or $(x_2, 0)$ (spiraling around them), except the stable manifold of the saddle, which separates the two basins.
- **(b)** $(x_1, 0)$ is a sink. At $(x_2, 0)$ the linearization has eigenvalues 0 and $-b$. On the center manifold $y \approx -P'(x)/b$, so $\dot{x} \approx -P'(x)/b \leq 0$ near x_2: points to the right of x_2 move toward it and points to the left move away, toward x_1. So $(x_2, 0)$ is a **saddle-node** (semistable). Orbits starting in the attracting sector on its right converge to $(x_2, 0)$, while all other orbits (everything on the other side of the strong stable manifold of $(x_2, 0)$) spiral into $(x_1, 0)$.

Exercise 7.12

For $\ddot{x} + \sin x = 0$, with $E = \frac{1}{2}\dot{x}^2 + 1 - \cos x$, the six qualitatively different solutions $x(t)$ are:

1. $x(t) \equiv 0$ (the stable equilibrium): a horizontal line.
2. Oscillation about 0 ($0 < E < 2$): a periodic wave with $|x| < \pi$ (a sinusoid-like curve whose period increases with amplitude).
3. $x(t) \equiv \pi$ (the unstable equilibrium; likewise $\equiv -\pi$): a horizontal line at height π.
4. The separatrix ($E = 2$, $\dot{x} > 0$): $x(t) = 4\arctan(e^t) - \pi$, an S-shaped curve increasing from $-\pi$ (as $t \to -\infty$) to π (as $t \to \infty$).
5. The other separatrix ($E = 2$, $\dot{x} < 0$): $x(t) = \pi - 4\arctan(e^t)$, decreasing from π to $-\pi$.
6. Rotation ($E > 2$): $x(t)$ increases monotonically without bound, with $x(t + T) = x(t) + 2\pi$, a staircase-like curve with periodically varying slope. (Rotation in the other direction is its mirror image $x \mapsto -x$.)

(For the separatrix: $\dot{x} = 2\cos(x/2)$ on the upper branch, which integrates to $\tan\frac{x+\pi}{4} = e^t$.)

Exercise 7.13

(a) $\dot{x} = x(3 - 3x - y)$, $\dot{y} = y(5 - 5y - 2x)$. The nullclines are $x = 0$ or $y = 3 - 3x$ for $\dot{x} = 0$, and $y = 0$ or $y = 1 - 0.4x$ for $\dot{y} = 0$. Equilibria: $(0, 0)$, $(1, 0)$, $(0, 1)$ and $(\frac{10}{13}, \frac{9}{13})$. With $J = \begin{pmatrix} 3 - 6x - y & -x \\ -2y & 5 - 10y - 2x \end{pmatrix}$:

- $(0, 0)$: eigenvalues $3, 5$, a source.
- $(1, 0)$: $\begin{pmatrix} -3 & -1 \\ 0 & 3 \end{pmatrix}$, eigenvalues $-3, 3$, a saddle.
- $(0, 1)$: $\begin{pmatrix} 2 & 0 \\ -2 & -5 \end{pmatrix}$, eigenvalues $2, -5$, a saddle.
- $(\frac{10}{13}, \frac{9}{13})$: $\begin{pmatrix} -30/13 & -10/13 \\ -18/13 & -45/13 \end{pmatrix}$, $\mathrm{tr} < 0$, $\det = \frac{1170}{169} > 0$ and $\mathrm{tr}^2 - 4\det > 0$, a stable node.

The two nullcline lines cross at the coexistence point. Above both lines, $\dot{x} < 0$ and $\dot{y} < 0$; below both, $\dot{x} > 0$ and $\dot{y} > 0$; between them, the flow points toward the crossing point (southeast for $x < \frac{10}{13}$ and northwest for $x > \frac{10}{13}$). Orbits stay in the positive quadrant and are bounded, so they are funneled into the coexistence node. **Outcome:** every initial condition with $x_0 > 0$ and $y_0 > 0$ tends to $(\frac{10}{13}, \frac{9}{13})$ (coexistence). The exceptions are the positive x-axis, whose points tend to $(1, 0)$, the positive y-axis, whose points tend to $(0, 1)$ (these axes are the stable manifolds of the saddles), and the origin itself.

(b) $\dot{x} = x(2 - 2x - y)$, $\dot{y} = y(1 - y - 3x)$. The nullclines are $y = 2 - 2x$ and $y = 1 - 3x$. They intersect at $x = -1$, so there is no interior equilibrium. Equilibria: $(0, 0)$, $(1, 0)$, $(0, 1)$. With $J = \begin{pmatrix} 2 - 4x - y & -x \\ -3y & 1 - 2y - 3x \end{pmatrix}$:

- $(0, 0)$: eigenvalues $2, 1$, a source.
- $(1, 0)$: $\begin{pmatrix} -2 & -1 \\ 0 & -2 \end{pmatrix}$, eigenvalue -2 (repeated), a sink.
- $(0, 1)$: $\begin{pmatrix} 1 & 0 \\ -3 & -1 \end{pmatrix}$, eigenvalues ± 1, a saddle.

For $x \geq 0$ the line $y = 2 - 2x$ lies above $y = 1 - 3x$. Above both, the flow points southwest; below both, northeast; between them, southeast. The strip between the lines is forward invariant (on $y = 2 - 2x$ the flow points down, and on $y = 1 - 3x$ it points right), and in it the flow is toward $(1, 0)$. **Outcome:** every initial condition with $x_0 > 0$ tends to $(1, 0)$ (species x survives, y dies out); initial conditions on the positive y-axis tend to $(0, 1)$.

Exercise 7.14

$\dot{x} = -z + \left(\frac{x^3}{3} - x\right)$, $\dot{z} = x$.

(a) System (7.51) is $\dot{x} = z - (\frac{x^3}{3} - x)$, $\dot{z} = -x$. Replacing t by $-t$ changes the sign of both right-hand sides and gives exactly the given system. So it is the Van der Pol system run backwards in time.

(b) $V = x^2 + z^2$ is positive definite, and

$$\dot{V} = 2x\left(-z + \tfrac{x^3}{3} - x\right) + 2zx = -2x^2 + \tfrac{2}{3}x^4 = \tfrac{2}{3}x^2(x^2 - 3).$$

So $\dot{V} \leq 0$ whenever $x^2 \leq 3$, and V is a Lyapunov function on the disk $x^2 + z^2 < 3$.

(c) On the open disk $x^2 + z^2 < 3$ we have $|x| < \sqrt{3}$, so $\dot{V} \leq 0$ and the disk is forward invariant (V cannot increase). The set $\{\dot{V} = 0\}$ in the disk is $\{x = 0\}$; an orbit staying on $x = 0$ has $\dot{x} = -z = 0$, so it is the origin. By Barbashin–LaSalle the disk is in the basin of attraction. So the largest open disk we can guarantee with this V has radius $\sqrt{3}$: $\{x^2 + z^2 < 3\}$.

(d) Reversing time, every solution of (7.51) starting in this disk (other than the origin) moves *away* from the origin as t increases, and tends to the origin as $t \to -\infty$. The origin is a source for (7.51), and no periodic orbit of (7.51) can lie inside the disk of radius $\sqrt{3}$; any periodic orbit must leave that disk.

Exercise 7.15

For $\dot{x} = 0$ ($x < 0$), $\dot{x} = \sqrt[3]{x}$ ($x \geq 0$), with $x(0) = 0$, there are (at least) two solutions: $x(t) \equiv 0$, and

$$x(t) = \begin{cases} 0, & t \leq 0, \\ \left(\frac{2t}{3}\right)^{3/2}, & t \geq 0. \end{cases}$$

Indeed for $t > 0$, $\dot{x} = \left(\frac{2t}{3}\right)^{1/2} = x^{1/3}$ (separating variables: $x^{-1/3}dx = dt$ gives $\frac{3}{2}x^{2/3} = t$). In fact, shifting gives a whole family: $x(t) = \left(\frac{2(t-s)}{3}\right)^{3/2}$ for $t \geq s$, 0 for $t < s$ ($s \geq 0$).

This does not contradict Theorem 7.14 because $f(x) = \sqrt[3]{x}$ (for $x \geq 0$) has $f'(x) = \frac{1}{3}x^{-2/3} \to \infty$ as $x \to 0^+$: f is not continuously differentiable (nor Lipschitz) near $x = 0$, so the hypotheses of the theorem fail.

Exercise 7.16

$\ddot{x} = x - ax^3$. Write $\dot{x} = y$, $\dot{y} = x - ax^3$, a conservative system with potential $P(x) = -\frac{1}{2}x^2 + \frac{a}{4}x^4$ and energy $E = \frac{1}{2}y^2 + P(x)$. Equilibria: $y = 0$ and $x(1 - ax^2) = 0$. The Jacobian is $\begin{pmatrix} 0 & 1 \\ 1 - 3ax^2 & 0 \end{pmatrix}$.

- **All a:** $(0, 0)$ has eigenvalues ± 1, a saddle, **unstable**.
- $a > 0$**:** also $(\pm 1/\sqrt{a},\, 0)$, where $1 - 3ax^2 = -2$ gives eigenvalues $\pm i\sqrt{2}$. These are centers (stable), the minima of the double-well potential. The phase plane has a figure-eight separatrix $E = 0$ through the origin enclosing two families of closed orbits (one around each center), and larger closed orbits outside the figure eight.
- $a = 0$**:** only the saddle; orbits are the hyperbolas $y^2 - x^2 = \text{const}$.
- $a < 0$**:** only the saddle at the origin. P is an inverted quartic, so orbits are hyperbola-like and unbounded (they escape to infinity, in finite time).

The only stable equilibria are $(\pm 1/\sqrt{a}, 0)$ for $a > 0$ (stable but not asymptotically stable); all other equilibria are unstable.

Exercise 7.17

$\dot{x}_1 = -3x_1$, $\dot{x}_2 = -8x_1^2 + 2x_2$.

(a) $x_1(t) = x_0 e^{-3t}$. Then $\dot{x}_2 - 2x_2 = -8x_0^2 e^{-6t}$; trying $x_2 = Ce^{-6t}$ gives $-8C = -8x_0^2$, so $C = x_0^2$. With $x_2(0) = y_0$:

$$x_1(t) = x_0 e^{-3t}, \qquad x_2(t) = x_0^2 e^{-6t} + (y_0 - x_0^2)e^{2t}.$$

(b) As $t \to \infty$, $x_1 \to 0$ always, and $x_2 \to 0$ iff the coefficient of e^{2t} vanishes. The stable manifold is the parabola $\{y_0 = x_0^2\}$, i.e. $x_2 = x_1^2$.

(c) As $t \to -\infty$, $x_1 = x_0 e^{-3t} \to 0$ iff $x_0 = 0$; then $x_2 = y_0 e^{2t} \to 0$. The unstable manifold is the x_2-axis $\{x_1 = 0\}$.

(d) As $t \to \infty$: $x_1 \to 0$; if $y_0 > x_0^2$ then $x_2 \to +\infty$; if $y_0 < x_0^2$ then $x_2 \to -\infty$; if $y_0 = x_0^2$ the orbit tends to the origin. As $t \to -\infty$: if $x_0 \neq 0$, $x_1 \to \pm\infty$ (the sign of x_0) and the e^{-6t} term dominates, so $x_2 \to +\infty$ with $x_2/x_1^2 \to 1$ (the orbit approaches the parabola $x_2 = x_1^2$ at infinity); if $x_0 = 0$ the orbit tends to the origin along the x_2-axis.

Exercise 7.18

$\dot{x} = -x^3 + x^2 y$, $\dot{y} = -y^3 - x^3$. The origin is the only equilibrium ($x^2(y - x) = 0$ and $y^3 = -x^3$ force $x = y = 0$). Take $V = x^2 + y^2$:

$$\dot{V} = 2x(-x^3 + x^2 y) + 2y(-y^3 - x^3) = -2x^4 - 2y^4 < 0 \quad \text{for } (x, y) \neq (0, 0).$$

V is positive definite and radially unbounded, so the origin is globally asymptotically stable. The solution starting at $(1, 2)$ exists for all $t \geq 0$, stays in $\{V \leq 5\}$, and tends to $(0, 0)$. Since $x^4 + y^4 \geq \frac{1}{2}V^2$, we get $\dot{V} \leq -V^2$, so $V(t) \leq \dfrac{5}{1 + 5t}$ and $|(x, y)| = O(t^{-1/2})$. The convergence is algebraic, not exponential, because the linearization at the origin is zero.